Data Mining: Concepts and Techniques (3rd Ed.)

Chapter 13 — Complete Q&A; Guide

Data Mining Trends and Research Frontiers

Jiawei Han, Micheline Kamber, Jian Pei | Morgan Kaufmann Publishers

2-Mark Short Answers • 6-Mark Long Answers • All Sections

Section	Topics
§ 13.1	Mining Complex Data Types: Time-Series, Graphs, Spatial, Web, Streams
§ 13.2	Other Methodologies: Statistical DM, Theoretical Foundations, Visual Mining
§ 13.3	Applications: Finance, Retail, Science, Intrusion Detection, Recommender Systems
§ 13.4	Data Mining and Society: Ubiquitous Mining, Privacy, Security
§ 13.5	Data Mining Trends and Research Frontiers

§ 13.1 — Mining Complex Data Types

Time-Series | Symbolic Sequences | Biological Sequences | Graphs | Spatial | Web | Streams

2-MARK QUESTIONS — §13.1

2 Marks

Q1. What is a time-series database? Give two examples.

A **time-series database** is a collection of sequences of numeric values obtained over repeated measurements at equal time intervals (e.g., per minute, per hour, per day). It captures temporal trends and patterns in data.

Examples: (1) Stock market prices — daily opening, closing, high, and low values for stocks over years. (2) Weather monitoring — hourly temperature, humidity, and wind speed readings from sensors worldwide.

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2 Marks

Q2. Distinguish between symbolic sequence data and biological sequence data.

Property	Symbolic Sequences	Biological Sequences
Nature	Ordered set of event or nominal data (not necessarily at equal intervals)	DNA/RNA nucleotides or amino acid sequences
Gap importance	Gaps between events usually do not matter much	Gaps are critical — insertions and deletions have biological meaning
Examples	Customer shopping sequences, web click streams	DNA sequences (ATCG), protein sequences (amino acids)
Length	Typically moderate	Usually very long with complex hidden semantics

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2 Marks

Q3. What are the four components of trend analysis in time-series data?

Trend analysis builds an integrated model using these four components: (1) **Trend/Long-term movements** — general direction of the time-series graph over time (moving averages, least squares). (2) **Cyclic movements** — long-term oscillations about a trend line. (3) **Seasonal variations** — recurring patterns in corresponding seasons of successive years (holidays). (4) **Random movements** — sporadic changes due to chance events like labour disputes or announcements.

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2 Marks

Q4. What is sequential pattern mining? Define a sequential pattern.

Sequential pattern mining discovers all frequent subsequences existing in one sequence or a set of sequences. A **sequential pattern** is a subsequence α that appears frequently in the data. A sequence α = ■a■a■...a■■ is a subsequence of β = ■b■b■...b■■ if there exist integers j■ < j■ < ... < j■ such that a■⊆b_j■, a■⊆b_j■, ..., a■⊆b_j■. For example, is a subsequence of .

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2 Marks

Q5. What is graph pattern mining? What are its main approaches?

Graph pattern mining discovers frequent subgraphs (subgraph patterns) in one or a set of graphs. The main approaches are: (1) **Apriori-based** — builds on the downward closure property (frequent subgraphs of a frequent graph are also frequent). (2) **Pattern growth-based** — grows patterns by extending one vertex/edge at a time. We can also mine only **closed graphs** where no proper supergraph carries the same support count.

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2 Marks

Q6. Distinguish between homogeneous and heterogeneous networks.

A **homogeneous network** has all nodes and links of the same type — for example, a friend network (person nodes, friend links) or a coauthor network. A **heterogeneous network** has nodes and links of different types — for example, a publication network linking authors, conferences, papers, and terms together, or a healthcare network linking doctors, patients, and diseases.

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2 Marks

Q7. What is web usage mining?

Web usage mining extracts useful information from server logs (e.g., user click streams). It discovers patterns related to user groups, understands user search patterns and associations, and predicts what users are looking for online. It helps improve search efficiency and effectiveness, and promotes products or information to the right users at the right time. Web search companies routinely conduct web usage mining to improve their quality of service.

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2 Marks

Q8. What are data streams? Why are they challenging to mine?

Data streams are data that flow into a system in vast volumes, change dynamically, are possibly infinite, and contain multidimensional features. Challenges: (1) They cannot be stored in traditional databases. (2) Systems can typically only read a stream once in sequential order. (3) The volume is too large for full offline analysis. Solutions include **sliding windows** (most recent data), **tilted-time windows** (recent data at fine granularity, old data at coarser granularity), and single-scan or few-scan approximation algorithms.

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2 Marks

Q9. What is subsequence matching in time-series data?

Subsequence matching in time-series data finds a set of sequences that contain subsequences similar to a given query sequence. Unlike normal database queries (exact match), similarity searches allow for slight differences. Dimensionality reduction techniques used include: (1) **Discrete Fourier Transform (DFT)** — transforms signals to frequency domain. (2) **Discrete Wavelet Transform (DWT)** — captures time-frequency information. (3) **Singular Value Decomposition (SVD)** — based on principal components analysis.

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2 Marks

Q10. What is a cyber-physical system (CPS)? Give an example.

A **cyber-physical system (CPS)** consists of a large number of interacting physical and information components that are interconnected to form large heterogeneous cyber-physical networks. **Example:** A patient care system linking a patient monitoring system (physical sensors) with a network of patient/medical information and an emergency handling system (cyber components). Other examples: transportation systems linking traffic cameras with control systems; battlefield command systems linking sensor networks with information analysis systems.

6-MARK QUESTIONS — §13.1

6 Marks

Q1. Explain mining sequence data — time-series, symbolic sequences, and biological sequences. Discuss methods for each type.

A **sequence** is an ordered list of events. There are three major types:

(a) Time-Series Data

Time-series data consist of long sequences of numeric values recorded at equal time intervals. Applications: stock market analysis, economic forecasting, weather monitoring, medical monitoring.

Key mining tasks:

- **Similarity search:** Find sequences differing slightly from a query. Uses DFT, DWT, or SVD for dimensionality reduction. Subsequence matching finds sequences containing similar subsequences.

- **Trend analysis:** Builds models using four components — trend movements (moving average/least squares), cyclic movements, seasonal variations (deseasonalised by autocorrelation), and random movements.
- **Forecasting:** ARIMA (auto-regressive integrated moving average), long-memory models, and autoregression predict future values.

(b) Symbolic Sequences

Consist of ordered sets of nominal events (not at equal intervals). Examples: customer shopping sequences, web click streams, program execution traces.

Key mining tasks:

- **Sequential pattern mining:** Discovers frequent subsequences. Methods include Apriori-based approaches and pattern growth (PrefixSpan). Can mine closed sequential patterns where no supersequence has the same support.
- **Constraint-based mining:** User constraints reduce search space — gap constraints (consecutive subsequences), periodic patterns (recurring in windows), and partial order patterns.
- **Sequence classification:** Feature-based (transform to vector), distance-based (using edit distance), and model-based (Hidden Markov Models) approaches.

(c) Biological Sequences

DNA/RNA nucleotide sequences and amino acid (protein) sequences. Very long with complex hidden semantics. Gaps (insertions, deletions) are biologically important.

Key mining tasks:

- **Sequence alignment:** Lines up sequences to find maximal identity. Pairwise alignment (two sequences) or multiple alignment. Local alignment (portions only) vs. global alignment (entire sequences). Goal: maximise alignment score counting matches as positive and gaps as negative.
- **Dynamic programming:** Commonly used for alignment. BLAST (Basic Local Alignment Search Tool) is one of the most popular tools.
- **Hidden Markov Models (HMM):** Represent statistical regularities of sequence classes. Forward algorithm (probability of sequence), Viterbi algorithm (most probable path), Baum-Welch algorithm (learns model parameters).

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Summary: Time-series → trend analysis, similarity search, ARIMA forecasting
Symbolic seq → sequential pattern mining, constraint-based, classification
Biological seq → alignment (BLAST), HMM, phylogenetic tree construction
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6 Marks

Q2. Explain the major themes in mining graphs and networks. Discuss statistical modeling, clustering, and role discovery in information networks.

Graph and network mining has become critical because graphs appear in social networks, biological networks, the Web, and many other domains. The major themes are:

1. Graph Pattern Mining

Discovers frequent subgraphs in a graph collection. Methods: Apriori-based (gSpan) and pattern growth-based (FFSM). Can mine closed graphs. Applications: graph indexing, similarity search, and graph classification using frequent discriminative subgraphs as features.

2. Statistical Modeling of Networks

Models homogeneous networks using generative models:

- **Erd■s-Rényi model (random graph):** Each edge exists independently with probability p

- **Watts-Strogatz model:** Captures small-world phenomenon — high local clustering with short global paths
- **Scale-free model:** Network follows power law (Pareto) distribution — few nodes have many connections (hubs)
- **Social network properties:** Densification power law (networks become denser over time), shrinking diameter, heavy-tailed degree distributions

3. Clustering and Classification

Uncovers hidden community structures, hubs, and outliers. Methods: partitioning, hierarchical, and density-based. Supervised classification and semi-supervised classification guide discovery with labelled data.

For **heterogeneous networks**: RankClus combines clustering and ranking mutually — highly ranked nodes contribute more to cluster cohesiveness; clusters help consolidate high ranking.

4. Role Discovery and Link Prediction

Discovers hidden roles (advisor-advisee, leader-follower) using expert constraints. Link prediction assesses ranking of expected relationships among candidate nodes — e.g., predicting which papers an author may write based on publication history.

Link mining tasks: link-based object classification, link type prediction, link existence prediction, link cardinality estimation, group detection, subgraph identification.

5. OLAP and Similarity Search in Networks

Path-based similarity considers different linkage paths for different similarity semantics. Hierarchical structures enable OLAP — drill down or dice information networks at different abstraction levels.

6. Network Evolution

Detecting evolving communities and anomalies helps predict trends and irregularities. Homogeneous networks evolve with same-type objects; heterogeneous networks reveal connected evolving objects of different types (authors, themes, venues).

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6 Marks

Q3. Explain mining spatial data, spatiotemporal data, multimedia data, text data, and web data. Give examples for each.

Beyond sequences and graphs, there are many other complex data types requiring specialised mining approaches:

1. Spatial Data Mining

Discovers patterns from geospatial data in vector, raster, or imagery formats. Topics: spatial association rules, co-location patterns (two spatial features frequently co-occur), spatial clustering, spatial classification, and spatial trend analysis.

> Example: Mining co-location patterns — hospitals and pharmacies frequently co-occur in urban areas. Spatial clustering groups similar weather stations. Spatial OLAP enables drill-down by region.

2. Spatiotemporal and Moving-Object Data

Spatiotemporal data relate to both space and time. Applications: discovering evolutionary history of cities, predicting earthquakes, global warming trends. Moving-object mining discovers:

- **Movement patterns:** moving clusters, leaders/followers, convoy (group moves together), swarm (periodic gathering), pincer (two groups converging)
- **Trajectory mining:** periodic patterns, trajectory clusters, outlier trajectories

> Example: GPS data from fleet vehicles → mine common routes, detect unusual detours, predict traffic jams. Wildlife GPS data → discover animal migration patterns and seasonal behaviours.

3. Multimedia Data Mining

Mines image, video, audio, and hypertext data. Integrates image processing, computer vision, data mining, and pattern recognition. Topics: content-based retrieval and similarity search, multimedia data cubes, classification and prediction, and association mining.

> Example: Mining an image database to find all photos with "mountain" scenery. Video mining: detecting scenes with specific objects. Audio mining: classifying music genres by spectral features.

4. Text Mining

An interdisciplinary field combining information retrieval, data mining, machine learning, statistics, and computational linguistics. Goal: derive high-quality information from text. Tasks:

- **Text categorisation:** classify news articles by topic (sports, finance, politics)
- **Text clustering:** group similar documents without labels
- **Concept/entity extraction:** identify named entities (people, organisations, locations)
- **Sentiment analysis:** determine opinion polarity (positive/negative) from product reviews
- **Document summarisation:** automatically generate concise summaries

5. Web Data Mining

Three main areas:

- **Web content mining:** Analyses text, multimedia, and structured data in web pages for keyword indexing, page ranking, entity resolution
- **Web structure mining:** Uses graph theory to analyse hyperlink structures — identifies authority pages, hubs, and web communities
- **Web usage mining:** Extracts patterns from server logs (click streams) to understand user navigation patterns and improve recommendations

§ 13.2 — Other Methodologies of Data Mining

Statistical Methods | Theoretical Foundations | Visual & Audio Data Mining

2-MARK QUESTIONS — §13.2

2 Marks

Q11. What is regression in statistical data mining? Give examples of regression types.

Regression methods predict the value of a response (dependent) variable from one or more predictor (independent) variables, where the variables are numeric. Types include: **Linear regression** (straight-line relationship), **Multiple regression** (several predictors), **Weighted regression** (different observation weights), **Polynomial regression** (curved relationship), **Nonparametric regression** (no assumed functional form), and **Robust regression** (handles significant outliers or non-normal errors).

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2 Marks

Q12. What is discriminant analysis? How does it differ from generalized linear models?

Discriminant analysis predicts a categorical response variable by determining discriminant functions (linear combinations of independent variables) that discriminate among groups. Unlike generalized linear models, it assumes that independent variables follow a **multivariate normal distribution**. Discriminant analysis attempts to find linear boundaries that maximally separate groups. It is commonly used in social sciences.

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2 Marks

Q13. What is the minimum description length (MDL) principle in data compression theory?

The **Minimum Description Length (MDL) principle** states that the best theory to infer from a data set is the one that minimizes the total length of: (1) the theory itself (encoded as bits) plus (2) the data when encoded using the theory as a predictor. In data mining, this means the best model is the one that most compactly encodes both the model structure and the data it describes. MDL naturally balances model complexity with goodness-of-fit, preventing overfitting.

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2 Marks

Q14. What is visual data mining? How does it differ from data visualization?

Visual data mining discovers implicit and useful knowledge from large data sets using data and knowledge visualization techniques. It combines data visualization with analytical mining capabilities. **Data visualization** simply presents data in visual form (charts, graphs, surfaces) for human inspection. Visual data mining goes further — it integrates the human's visual processing power with automated mining algorithms to find patterns. Types: data visualization, mining result visualization, mining process visualization, and interactive visual mining.

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2 Marks

Q15. What is audio data mining? What advantage does it offer over visual data mining?

Audio data mining uses audio signals (pitch, rhythm, tune, melody) to indicate the patterns of data or features of data mining results. Instead of watching visual displays, users listen for interesting or unusual patterns. **Advantage over visual mining:** Visual mining requires constant concentration on watching patterns — tiring over long periods. Audio mining allows users to detect interesting or unusual patterns more naturally and with less visual fatigue, since the auditory system can process sound passively. It is an interesting complement to visual data mining.

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2 Marks

Q16. What is the microeconomic view of data mining?

The **microeconomic view** considers data mining as the task of finding patterns that are interesting only to the extent that they can be used in the **decision-making process** of an enterprise (e.g., marketing strategies, production plans). Patterns are considered interesting only if they are actionable. Enterprises are seen as facing optimization problems where the goal is to maximize the utility or value of a decision. Under this view, data mining becomes a nonlinear optimization problem where patterns are evaluated by their economic utility.

6-MARK QUESTIONS — §13.2

6 Marks

Q4. Explain the major statistical data mining methods and their applications in data analysis.

Statistical data mining draws from well-established statistical methods applicable to scientific, economic, and social data. Major methods:

Statistical Method	Description	Application Example
Regression	Predicts numeric response from predictors. Types: linear, multiple, weighted, polynomial, robust	Predicting house prices from size, location, age
Generalized Linear Models (GLM)	Allows categorical response; includes logistic and Poisson regression	Predicting probability of loan default (logistic)
Analysis of Variance (ANOVA)	Compares k population means; determines if at least two differ significantly	Testing if three fertilizers yield different crop outputs
Mixed-Effect Models	Analyses grouped data with response variable related to covariates in groups	Longitudinal medical study with multiple patient visits
Factor Analysis	Determines which variables generate a given unobservable factor	Finding intelligence factors from student test scores
Discriminant Analysis	Predicts categorical response; assumes multivariate normal distribution	Classifying tumours as malignant or benign
Survival Analysis	Predicts probability of surviving to time t; Kaplan-Meier, Cox models	Estimating patient survival after cancer treatment
Quality Control	Shewhart/CUSUM charts monitor group summary statistics	Statistical process control in manufacturing

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Key difference from data mining: Statistical methods: typically designed for
small, homogeneous, numeric data sets Data mining: designed for huge,
multidimensional, possibly complex data types Both are complementary — data mining
can automate and scale statistical methods
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6 Marks

Q5. Discuss the theoretical foundations of data mining. Explain five proposed theories with examples.

A solid theoretical foundation is critical for a coherent framework for developing, evaluating, and practising data mining. Five major theories:

1. Data Reduction Theory

Data mining reduces data representation — trades accuracy for speed on very large databases. Techniques: SVD (principal components), wavelets, regression, log-linear models, histograms, clustering, sampling, and index trees. Goal: obtain quick approximate answers to queries.

2. Data Compression Theory

Data mining compresses data by encoding using association rules, decision trees, clusters, or bits. Based on the Minimum Description Length (MDL) principle — the best model minimizes the combined length of the model description plus the data encoded using that model.

3. Probability and Statistical Theory

Data mining discovers joint probability distributions of random variables. Examples: Bayesian belief networks model conditional independence relationships, hierarchical Bayesian models capture latent structures. This theory provides a principled framework for uncertainty quantification.

4. Microeconomic View

Patterns are interesting only if they can be acted on in enterprise decision-making. Treats data mining as a nonlinear optimization problem where the objective is to maximize utility/value of discovered patterns. Considers enterprises as facing optimization challenges in marketing, production, and logistics.

5. Pattern Discovery and Inductive Databases

Data mining discovers patterns (associations, classifications, sequential patterns, clusters) occurring in data. A knowledge base can be viewed as an inductive database of data and patterns. Users interact with the system by querying both the data and the patterns.

Requirements for an ideal framework: (1) Models typical data mining tasks (association, classification, clustering). (2) Has probabilistic nature. (3) Handles different forms of data. (4) Considers iterative and interactive essence of data mining. None of the current theories fully satisfies all four — further research is ongoing.

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6 Marks

Q6. Explain visual data mining in detail. Describe the four ways data visualization and data mining can be integrated.

Visual data mining discovers implicit and useful knowledge from large data sets by leveraging the power of the human visual system (eyes + brain as a parallel processing engine with a large knowledge base). It combines two disciplines: data visualization and data mining.

Four Integration Approaches:

1. Data Visualization

Data is displayed at different granularity or abstraction levels, or as different combinations of attributes. Forms include: boxplots (show statistical distributions), 3D data cubes, distribution charts, scatter plots, surfaces, and link graphs. Gives users a clear impression of data characteristics in large datasets.

> Example (StatSoft): Boxplots showing multiple variable combinations side by side, or multidimensional data distribution in 3D scatter plots. Users can visually identify outliers and cluster tendencies.

2. Data Mining Result Visualization

Presents results or knowledge obtained from data mining in visual forms:

- Decision trees — hierarchical tree structure showing split conditions (MineSet visualization)
- Association rules — plane with pillars representing rules and their strengths
- Clusters — coloured groupings showing cluster properties (IBM Intelligent Miner)

- Scatter plots — results from SAS Enterprise Miner for classification

3. Data Mining Process Visualization

Shows various mining processes visually — data extraction sources, preprocessing, cleaning, integration, mining method selection, result storage, and viewing. Example: Clementine data mining system shows complete data flow in a visual pipeline. Users see exactly which database is queried, how data flows through preprocessing steps, and how results are generated.

4. Interactive Visual Data Mining

Visualization tools are used during the mining process to help users make smart mining decisions. Example: Perception-Based Classification (PBC) system from University of Munich — data distribution displayed using coloured sectors, helping users decide which sector to split first and where to place split points. User can guide the mining process interactively.

Audio Data Mining: Maps data patterns to sound — pitch, rhythm, tune, and melody indicate different pattern features. Advantages: (1) Less visual fatigue — users can listen while doing other tasks. (2) Natural anomaly detection — unusual sounds stand out immediately. (3) Complements visual mining for a richer mining experience.

§ 13.3 — Data Mining Applications

Finance | Retail & Telecom | Science & Engineering | Intrusion Detection | Recommender Systems

2-MARK QUESTIONS — §13.3

2 Marks

Q17. How is data mining used for detecting money laundering?

To detect money laundering and other financial crimes, multiple heterogeneous databases (bank transactions, federal/state crime histories) are integrated. Then data analysis tools are applied: **data visualization tools** display unusual transaction activity by time/customer groups; **link analysis tools** identify suspicious relationships between customers; **outlier analysis** detects unusual fund transfer amounts; **sequential pattern analysis** characterises unusual access sequences. These tools help investigators focus on suspicious cases for detailed examination.

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2 Marks

Q18. What is customer retention analysis in retail data mining?

Customer retention analysis uses loyalty card data to register sequences of purchases by individual customers over time. By applying **sequential pattern mining**, changes in customer consumption and loyalty are investigated. The results can suggest adjustments to pricing and variety of goods to retain existing customers and attract new ones. It helps retailers understand which product categories customers are loyal to and which they are dropping, enabling proactive marketing interventions.

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2 Marks

Q19. What is signature-based detection in intrusion detection systems?

Signature-based detection uses attack patterns (signatures) that are pre-configured and predetermined by domain experts. The IDS/IPS monitors network traffic for matches to these patterns and reports/blocks detected matches. **Limitations:** (1) Signatures must be updated laboriously whenever new software versions or configurations change. (2) It can only identify attacks that match known signatures — cannot detect novel or previously unknown intrusions. Signature-based systems are complemented by anomaly-based detection for comprehensive coverage.

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2 Marks

Q20. What is a recommender system? Name two approaches used.

A **recommender system** helps consumers find products/services (books, movies, restaurants, news articles) that are likely to be of interest to them, from among millions of options. Two main approaches: (1) **Content-based approach** — recommends items similar to those the user preferred in the past, based on product features and textual descriptions. (2) **Collaborative filtering approach** — recommends items based on opinions of other customers who have similar tastes or preferences. Hybrid systems combine both.

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2 Marks

Q21. Distinguish between memory-based and model-based collaborative recommender systems.

Memory-based methods use heuristics to estimate ratings from the entire collection of previously rated items. They use k-nearest-neighbor (k-NN) to find k users most similar to the target user (using Pearson correlation or cosine similarity), then compute a weighted aggregate of their ratings. **Model-based methods** use the collection of ratings to learn a model (e.g., probabilistic models, Bayesian networks, clustering), which is then used for predictions. Model-based methods generally scale better to large datasets but require more computation during training.

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2 Marks

Q22. What shift has occurred in scientific data analysis due to massive data collection?

Historically, scientific data analysis followed a "formulate hypothesis → build model → evaluate results" paradigm with relatively small, homogeneous data sets. Due to massive data collection technologies (telescopes, satellite sensors, GPS, biological analysis tools), science has shifted to a "collect and store data → mine for new hypotheses → confirm with data or experimentation" process. This shift requires data mining to handle huge volumes of high-dimensional, heterogeneous, stream data containing rich spatial and temporal information — a fundamentally different challenge.

6-MARK QUESTIONS — §13.3

6 Marks

Q7. Discuss data mining applications in financial data analysis with detailed examples.

Financial data is often complete, reliable, and of high quality — ideal for systematic data mining. Major applications:

1. Data Warehouses for Multidimensional Analysis

Financial data warehouses support analysis along dimensions like time, region, and sector. A financial officer can view debt and revenue changes by month/region with statistics (max, min, total, average, trend, standard deviation). Advanced data cube concepts — multifeature cubes, discovery-driven cubes, regression cubes — all play important roles.

2. Loan Payment Prediction and Credit Analysis

Critical for bank operations. Data mining tasks: attribute selection and relevance ranking to identify critical factors (loan-to-value ratio, debt ratio, payment-to-income ratio, credit history, income, education). Analysis reveals dominant factors — e.g., payment-to-income ratio may be dominant while education level is not. Banks can then revise loan-granting policies accordingly.

> Example: A bank finds that customers with payment-to-income ratio < 35% have 95% repayment rate. It revises policy to focus on this factor, previously denying loans to some low-risk customers.

3. Customer Classification and Clustering for Targeted Marketing

Classification identifies the most crucial factors influencing customer banking decisions. Clustering identifies groups of customers with similar loan payment behaviours, enabling targeted marketing campaigns. Example: multidimensional clustering groups customers by spending patterns for personalized credit card product offers.

4. Detection of Money Laundering and Financial Crimes

Integration of multiple heterogeneous databases (bank transaction + crime history). Mining tasks:

- **Data visualization:** Display transaction activities by time/customer groups — detect large cash flows
- **Link analysis:** Identify relationships among different customers and suspicious activities
- **Clustering tools:** Group similar cases together for comparison
- **Outlier analysis:** Detect unusual fund transfers or atypical transaction amounts
- **Sequential pattern analysis:** Characterise unusual access sequences or transaction patterns

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6 Marks

Q8. Explain data mining for intrusion detection and prevention systems. Discuss how data mining methods enhance IDS performance.

An **intrusion** is any set of actions threatening the integrity, confidentiality, or availability of a network resource. **Intrusion Detection Systems (IDS)** monitor and report malicious activity. **Intrusion Prevention Systems (IPS)** actively block detected intrusions. Two main detection approaches:

Detection Approaches:

Approach	How it works	Advantage	Disadvantage
Signature-based	Matches network traffic against pre-configured attack patterns (signatures)	Low false positive rate; fast detection of known attacks	Cannot detect novel attacks; signatures need constant updates
Anomaly-based	Builds models of normal behaviour (profiles); detects significant deviations	Detects novel, previously unknown intrusions	High false positive rate; human analyst needed to verify

How Data Mining Enhances IDS:

1. New data mining algorithms: For signature-based detection, training data are labelled "normal/intrusion" → train classifier. For anomaly detection: clustering and outlier analysis build normal behaviour models.

2. Association, correlation, and discriminative patterns: Find relationships between system attributes → identify useful attributes for intrusion detection → new derived attributes (aggregate counts of traffic matching a pattern) improve classifiers.

3. Stream data analysis: Intrusions are transient and dynamic — real-time stream mining is essential. Sequential pattern mining identifies malicious event sequences. Dynamic classification models adapt to evolving attack patterns.

4. Distributed data mining: Intrusions launched from multiple locations need analysis of data from several network locations simultaneously → distributed mining detects coordinated attacks.

5. Visualization and querying tools: Security analysts need visual dashboards showing associations, clusters, and outliers. Graphical interfaces allow querying of intrusion detection results and network data.

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6 Marks

Q9. Explain recommender systems in detail. Discuss content-based, collaborative, and hybrid approaches. What challenges do they face?

Recommender systems help consumers find products of interest from millions of options (books, movies, restaurants, news). They provide **personalisation** — "a personalized store for every customer" (Amazon).

1. Content-Based Approach

Recommends items similar to items the user preferred or queried in the past. Relies on product features and textual descriptions. For movies: similar genres, directors, actors. For articles: similar terms. User profiles are obtained explicitly (questionnaires) or learned from transaction behaviour.

Limitation: Recommendations are limited by the features used to describe items. Cannot discover serendipitous recommendations (something completely different but interesting).

2. Collaborative Filtering Approach

Predicts utility of items for user u based on ratings by other similar users.

- **Memory-based:** Uses entire collection of past ratings. Finds k-NN (k nearest neighbours) of target user using Pearson correlation or cosine similarity. Weighted aggregate of neighbours' ratings gives prediction.
- **Model-based:** Learns a model (probabilistic, Bayesian networks, clustering) from ratings. Predictions made from model — scales better to large datasets.

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Recommendation problem formulation: Let C = set of users, S = set of items Utility function u: C × S → rating (initially defined only for rated items) Goal: extrapolate from known to unknown ratings Recommend items with highest predicted utility/rating to each user
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3. Hybrid Approach

Integrates content-based and collaborative methods. The Netflix Prize ($1,000,000 competition) showed that blending multiple substantially different predictors gives much better accuracy than refining a single technique. Ensemble methods combine many models.

Challenges:

- **Scalability:** Systems must search millions of neighbours in real time — requires efficient similarity computation
- **Cold start problem:** New users with no history cannot receive good recommendations
- **False positives:** Recommending items users dislike damages trust
- **Feature limitation:** Content-based systems are limited by available item features
- **Sparsity:** Most users rate only a tiny fraction of items — rating matrix is very sparse

§ 13.4 — Data Mining and Society

Ubiquitous Mining | Privacy-Preserving Data Mining | Security and Social Impacts

2-MARK QUESTIONS — §13.4

2 Marks

Q23. What is ubiquitous data mining? Give two examples from daily life.

Ubiquitous data mining is the constant, ever-present application of data mining in many aspects of daily life — often without the user's awareness. Examples: (1) **Credit card fraud detection** — companies analyse spending patterns in real time and call customers about suspicious charges, saving billions annually. (2) **Email spam filtering** — classification algorithms automatically identify and delete spam before users see it. (3) **Google search personalisation** — PageRank and click-stream mining tailor search results and advertisements to user interests.

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2 Marks

Q24. What is privacy-preserving data mining? What is the fundamental trade-off?

Privacy-preserving data mining (PPDM) deals with obtaining valid data mining results without disclosing the underlying sensitive data values. Also called privacy-enhanced or privacy-sensitive data mining. The **fundamental trade-off:** Protecting privacy requires reducing the granularity of data representation (generalising from individuals to groups), which causes **information loss** and potentially reduces the usefulness of mining results. Every privacy protection technique must balance this trade-off between information preservation and privacy protection.

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2 Marks

Q25. Explain the k-anonymity method in privacy-preserving data mining.

The **k-anonymity method** alters individual records so that each record maps onto at least k other records in the data — making it impossible to uniquely identify an individual among a group of at least k people. Techniques used: **generalisation** (replacing specific values with broader categories — age "28" → "20-30") and **suppression** (removing certain attribute values entirely). **Weakness:** If all k records sharing a quasi-identifier have the same sensitive attribute value, the sensitive value can still be inferred — handled by the l-diversity extension.

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2 Marks

Q26. What is the l-diversity model and how does it improve on k-anonymity?

The **l-diversity model** enforces **intragroup diversity** of sensitive attribute values within each group of k records. While k-anonymity ensures a record cannot be uniquely identified, it fails when all k records in a group share the same sensitive value (e.g., all have the same disease). The l-diversity model requires that each equivalence class has at least l well-represented values for the sensitive attribute, making it sufficiently difficult for adversaries to use combinations of attributes to infer sensitive values even within a group.

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2 Marks

Q27. What is differential privacy? What guarantee does it provide?

Differential privacy guarantees that for any two data sets differing only on a tiny portion (e.g., a single individual's record), a differentially private algorithm behaves approximately the same on both data sets. This means the presence or absence of any individual's data cannot significantly affect the query output — protecting individual privacy. Differentially private algorithms add carefully calibrated noise to query results, ensuring mathematical privacy guarantees. Research in differential privacy for data mining is ongoing and rapidly expanding.

6-MARK QUESTIONS — §13.4

6 Marks

Q10. Discuss privacy-preserving data mining methods in detail. Explain randomisation, k-anonymity, l-diversity, distributed privacy, and downgrading methods.

As data mining tools grow more powerful and personal data becomes more accessible, protecting individual privacy becomes critical. Privacy-preserving data mining (PPDM) aims to mine valid results without disclosing sensitive data. Major methods:

1. Randomisation Methods

Add noise to data to mask sensitive attribute values. The noise must be: (a) sufficiently large to prevent recovery of individual values; (b) small enough that aggregate distributions and mining results are still meaningful.

```
Example: Attribute "salary" with actual value $85,000 Add random noise: $85,000 +
random(-10000, +10000) = $91,234 (published) Individual record privacy: attacker
cannot recover exact salary Aggregate result preserved: distribution of salaries
approximately maintained
```

2. k-Anonymity Method

Reduces granularity so any record matches at least k other records. Uses generalisation (specific → range) and suppression (remove identifying attributes).

k-Anonymity Example	Before	After (k=2)
Age	28	20-30
Zip code	12345	1234*
Gender	Male	Male
Disease (sensitive)	Cancer	Cancer (not generalised)

3. l-Diversity Model

Extends k-anonymity by requiring at least l distinct values for sensitive attributes within each group. Prevents inference attacks when all k records share the same sensitive value. Makes it hard for adversaries to determine an individual's sensitive value even within an anonymised group.

4. Distributed Privacy Preservation

Large data sets partitioned and distributed across sites:

- **Horizontally partitioned:** Each site holds a subset of records (different customers at different banks)
- **Vertically partitioned:** Each site holds a subset of attributes (different attributes at different companies)

Sites share only limited aggregate information using protocols — individual records never leave their site. Mining result quality is maintained while individual site data remains private.

5. Downgrading Data Mining Results

Even when data is not shared, mining outputs (association rules, classification models) may reveal sensitive patterns. Solution: modify data or mining results — hiding some association rules, slightly distorting classification models — to prevent privacy violations without exposing raw data.

Differential Privacy (emerging approach): Mathematical guarantee that algorithm output is nearly identical whether or not any individual's data is included. Achieved by adding calibrated noise. Provides rigorous, quantifiable privacy guarantees.

6 Marks

Q11. Explain the social impacts of data mining — both benefits and concerns. Discuss ubiquitous mining and invisible data mining with examples.

Data mining permeates modern society, bringing both significant benefits and raising important social concerns:

Benefits of Ubiquitous Data Mining:

Area	Data Mining Application	Benefit
Retail (Wal-Mart)	Suppliers analyse buying patterns by store; control inventory	Optimal product placement; reduced waste
Online shopping (Amazon)	Collaborative recommender systems; personalised store for each customer	Cross-selling; customer retention; increased sales
Credit card companies	Fraud detection by mining spending patterns in real time	Billions saved annually; customer protection
Internet search (Google)	PageRank; click-stream mining; personalised ads and search results	Relevant results; targeted advertising; cost savings
Email systems	Spam classification using machine learning classifiers	Billions of spam emails blocked daily
CRM systems	Study browsing/purchasing patterns to personalise ads	Less annoying mass marketing; higher satisfaction

Invisible Data Mining:

"Smart" software — search engines, customer-adaptive web services, email managers, ticket masters — incorporates data mining into functional components, often without the user's knowledge. Google interpreting "Boston New York" as transit schedules but "Boston Paris" as flight schedules is a result of invisible frequent pattern mining of previous click streams.

Privacy and Security Concerns:

- Personal data (credit card records, health records, financial records, biological traits) may be mined without consent
- Data mining can disclose information about individuals even when only aggregate results are published
- Identity theft, targeted surveillance, and discriminatory profiling are potential misuses
- Even when identifying information is removed, re-identification is often possible by combining attributes

Balance: Most data mining does not involve personal data (natural resources, meteorology, astronomy, biology). The real risk is unrestricted access to individual records. Collaboration of computer scientists, policy experts, social scientists, lawyers, and consumers is needed to build robust privacy and security mechanisms.

§ 13.5 — Data Mining Trends and Research Frontiers

Current Trends | Emerging Research Areas | Future Directions

2-MARK QUESTIONS — §13.5

2 Marks

Q28. What does "constraint-based mining" mean and why is it important for scalable data mining?

Constraint-based mining allows users to specify constraints that guide data mining systems in their search for interesting patterns and knowledge. This provides: (1) **Added user control** — users define what they want, reducing irrelevant pattern discovery. (2) **Search space pruning** — constraints eliminate large portions of the hypothesis space before mining begins, dramatically improving efficiency. (3) **Better scalability** — by reducing the problem size, it makes mining feasible on huge data sets. Example: constraint "only find itemsets containing products costing > $100" reduces search space significantly.

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2 Marks

Q29. Why is integration of data mining with cloud computing systems important?

Integration with **cloud computing** is important because: (1) **Data availability** — cloud stores massive distributed data sets accessible from anywhere. (2) **Scalability** — cloud provides elastic computation resources to scale mining tasks dynamically. (3) **High performance** — distributed cloud architectures enable parallel mining on petabyte-scale data. (4) **Portability** — mining subsystems run as cloud services accessible from multiple platforms. Integration ensures a seamless, unified framework for multidimensional data analysis and exploration at any scale.

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2 Marks

Q30. What are the major challenges in biological data mining?

Biological data presents unique challenges: (1) **Complexity:** DNA/protein sequences carry complicated, hidden semantic meaning requiring domain knowledge. (2) **High dimensionality:** microarray data has thousands of genes as dimensions but few samples. (3) **Heterogeneity:** integrating diverse biological data types (genomic, proteomic, clinical, pathway data) from multiple sources. (4) **Scale:** biological databases are massive and growing. (5) **Specialised mining needed:** standard association/classification methods don't directly apply — need algorithms for sequence alignment, pathway analysis, and biological network mining.

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2 Marks

Q31. What is the role of data mining in software and system engineering?

Data mining in **software/system engineering** can: (1) **Monitor system status** and improve performance by analysing execution traces. (2) **Isolate software bugs** — mining execution traces of buggy programs reveals patterns and outliers pointing to bug locations (essentially a data mining process). (3) **Detect software plagiarism** by comparing code patterns. (4) **Analyse computer faults** and system malfunctions. (5) **Uncover network intrusions**. Data mining for software engineering can operate on static (dump traces) or dynamic (real-time stream) data.

6-MARK QUESTIONS — §13.5

6 Marks

Q12. Describe the major current trends in data mining research and development. Explain at least eight distinct trends.

Data mining is a rapidly evolving field. Key trends shaping its future:

Trend	Description	Key Research Challenges
Application exploration	Expansion from business to science, government, biomedicine, counterterrorism, mobile mining. Trend toward application-specific and invisible mining tools.	Domain knowledge integration; real-time mining; task-specific algorithms
Scalable and interactive mining	Constraint-based mining provides added user control while improving efficiency. Single-scan algorithms for huge data sets.	Handling terabyte/petabyte scales; sub-linear time algorithms
Integration with search engines and databases	Data mining as an embedded component of DB systems, warehouses, cloud computing, and search engines. Tightly coupled, not standalone.	API design; query integration; semantic integration
Mining social and information networks	Critical due to ubiquity of social networks, heterogeneous networks, link prediction, community evolution.	Scalability for billion-node graphs; handling heterogeneity
Spatiotemporal and CPS mining	GPS, sensors, mobile phones generate massive spatiotemporal data. Cyber-physical systems require real-time mining.	Real-time spatial queries; uncertainty handling; energy efficiency
Multimedia, text, web mining	Mining unstructured/semi-structured data. Sentiment analysis, topic modeling, deep web mining, multimedia retrieval.	Semantic understanding; multilingual mining; visual comprehension
Biological and biomedical mining	Mining DNA/protein sequences, microarray data, biological pathways, biomedical literature.	Multi-omics integration; knowledge transfer from model organisms
Distributed and real-time stream mining	Internet, cloud, sensor networks need distributed algorithms. Stock analysis, intrusion detection, mobile data require real-time models.	Communication overhead; concept drift; limited memory streams
Privacy and security	Growing concern as powerful mining tools access more personal data. Need for differential privacy, k-anonymity, secure multi-party computation.	Balancing utility vs. privacy; regulation compliance
Visual and audio mining	Integrating human visual/auditory systems with mining. Interactive visual mining helps users guide the discovery process.	Scalable visualisation for complex data; audio pattern encoding

6 Marks

Q13. Write comprehensive notes on: (a) Mining biological data, (b) Mining social networks, (c) Privacy in data mining.

(a) Mining Biological and Biomedical Data

The unique combination of complexity, richness, size, and importance makes biological data a critical domain for data mining.

Key mining tasks:

- **DNA/protein sequence mining:** Alignment (BLAST), motif discovery, phylogenetic tree construction, gene function prediction
- **High-dimensional microarray data:** Gene expression profiling with thousands of genes — requires dimensionality reduction and biclustering

- **Biological pathway analysis:** Discovering regulatory networks — which genes activate/inhibit other genes. Finding regulatory sequences in DNA that control gene expression
- **Network analysis:** Protein-protein interaction networks, metabolic networks, gene regulatory networks
- **Biomedical literature mining:** Text mining of PubMed articles to extract drug-disease relationships, gene functions

> Example: Mining DNA regulatory sequences Goal: identify DNA segments ("regulatory sequences") controlling specific gene expression Method: mine large biological data sets for patterns correlating with gene activation/suppression Result: Discover transcription factor binding sites → understand disease mechanisms → drug targets

(b) Mining Social and Information Networks

Social networks (Facebook, Twitter, LinkedIn) and information networks (citation networks, web graphs) contain rich structural information.

Key mining tasks:

- **Community detection:** Find cohesive subgroups — friend circles, research communities, terrorist cells
- **Link prediction:** Predict future connections — friend recommendations, citation predictions
- **Influence propagation:** Model how information, opinions, or diseases spread through networks
- **Heterogeneous network mining:** Mine networks with multiple node/link types — RankClus combines ranking and clustering
- **Network evolution:** Track evolving communities, detect structural changes, predict network growth
- **Small-world and scale-free properties:** Most social networks exhibit power-law degree distribution (few hubs, many peripheral nodes) and small-world phenomenon (short paths between any two nodes)

(c) Privacy in Data Mining

As data mining grows more powerful and personal data more accessible, privacy protection becomes increasingly critical.

Privacy concern types:

- **Access to sensitive records:** Credit card transactions, health records, financial data, criminal records
- **Re-identification attacks:** Combining non-sensitive attributes (ZIP code + gender + age) uniquely identifies individuals
- **Inference attacks:** Mining results (association rules) reveal sensitive information not directly in data

Protection methods: Randomisation (add noise), k-anonymity (generalise to groups), l-diversity (ensure sensitive attribute diversity), distributed privacy (data never leaves sites), differential privacy (mathematical guarantees), and downgrading mining results (hide sensitive rules).

Path forward: Collaboration among technologists, social scientists, lawyers, governments, and companies is needed to develop rigorous privacy and security protection mechanisms for data publishing and mining — allowing us to continue reaping data mining benefits while protecting individual rights.

Quick Reference — Chapter 13 Key Terms and Definitions

Term	Definition
Time-series data	Sequence of numeric values at equal time intervals; e.g., stock prices, temperature readings
Subsequence matching	Finding sequences containing subsequences similar to a given query
ARIMA	Auto-Regressive Integrated Moving Average — time series forecasting method
Sequential pattern	Frequent subsequence in one or more sequences; mined by PrefixSpan, SPADE
Closed sequential pattern	Sequential pattern with no supersequence having the same support
Sequence alignment	Lining up biological sequences to achieve maximum identity; pairwise or multiple
BLAST	Basic Local Alignment Search Tool — most popular biological sequence analysis tool
Hidden Markov Model	Probabilistic model for biological sequences; uses Forward, Viterbi, Baum-Welch algorithms
Graph pattern mining	Discovering frequent subgraphs; Apriori-based or pattern growth approaches
Scale-free network	Network following power-law degree distribution; few hubs with many connections
Small-world phenomenon	High local clustering + short global path length; found in most social networks
RankClus	Algorithm combining ranking and clustering for heterogeneous information networks
Web content mining	Analysing web page content for indexing, entity resolution, page ranking
Web structure mining	Graph analysis of hyperlink structures to find authority pages and hubs
Web usage mining	Extracting patterns from server logs (click streams) to understand user behaviour
Data stream	Vast volumes of data flowing in dynamically, possibly infinite, read once
CPS (Cyber-Physical System)	Interacting physical and information components forming heterogeneous networks
k-anonymity	Each record maps to at least k other records; uses generalisation and suppression
l-diversity	Extension of k-anonymity; enforces diversity of sensitive values within each group
Differential privacy	Algorithm behaves similarly on data sets differing by one individual — strong mathematical guarantee
Privacy-preserving DM	Obtaining valid mining results without disclosing sensitive data values
Recommender system	System predicting and recommending items of interest to users from large catalogues
Collaborative filtering	Recommending based on preferences of similar users; memory-based or model-based

Anomaly-based IDS	Intrusion detection by modelling normal behaviour and flagging significant deviations
Signature-based IDS	Intrusion detection by matching network traffic against known attack patterns
Ubiquitous data mining	Constant presence of data mining in daily life — shopping, searching, email
Invisible data mining	Mining embedded in software (search engines, email) without user awareness
Trend analysis components	Long-term trend + cyclic movements + seasonal variations + random movements
MDL principle	Best model minimizes combined length of model + data encoded using the model
Audio data mining	Using sound signals (pitch, rhythm, tune) to represent data patterns and mining results